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Boundary & Top View

Binary Trees, Boundary Traversal, Top View · Mentor-Led Lab (Student Edition)

Developed by TalenciaGlobal

Difficulty ★★★★ Advanced

Problem

Trace the outline of a binary tree. Produce the anti-clockwise boundary traversal - the root, then the left boundary downward, then all the leaves left to right, then the right boundary upward - and count how many columns the top view spans.

Implement the boundary traversal and the column count from scratch.

What to Compute

- Boundary: the anti-clockwise boundary - root, left boundary (top-down, excluding leaves), all leaves (left to right), right boundary (bottom-up, excluding leaves).
- Top View: the number of distinct horizontal columns (root = 0, left = -1, right = +1).

Input / Output Format

Input: One line: the tree in level order (values and N).

Output: Two lines: Boundary and Top View.

Examples

Example 1

Input: tree = 20 8 22 4 12 N 25 N N 10 14

Output:

```
Boundary: 20 8 4 10 14 25 22
Top View: 5
```

Explanation: Root 20, then left boundary 8, then leaves 4, 10, 14, 25, then right boundary 22 (bottom-up); the nodes occupy 5 distinct columns.

Example 2

Input: tree = 1 2 3 N N 4 5

Output:

```
Boundary: 1 2 4 5 3
```

```
Top View: 4
```

Explanation: Root 1, left boundary contributes nothing extra (2 is a leaf), leaves are 2, 4, 5, then right boundary 3; the tree spans 4 columns.

Constraints

- The root is printed once; it is not repeated among the leaves.
- Left and right boundaries exclude leaf nodes (leaves are listed in the leaves pass).
- A single column index is assigned per node: root 0, left child -1, right child +1.

Sample Run

Sample Input

```
20 8 22 4 12 N 25 N N 10 14
```

Sample Output

```
Boundary: 20 8 4 10 14 25 22
Top View: 5
```